

GEOGRAPHY

INDIAN GEOGRAPHY



MINERALS & INDUSTRIES

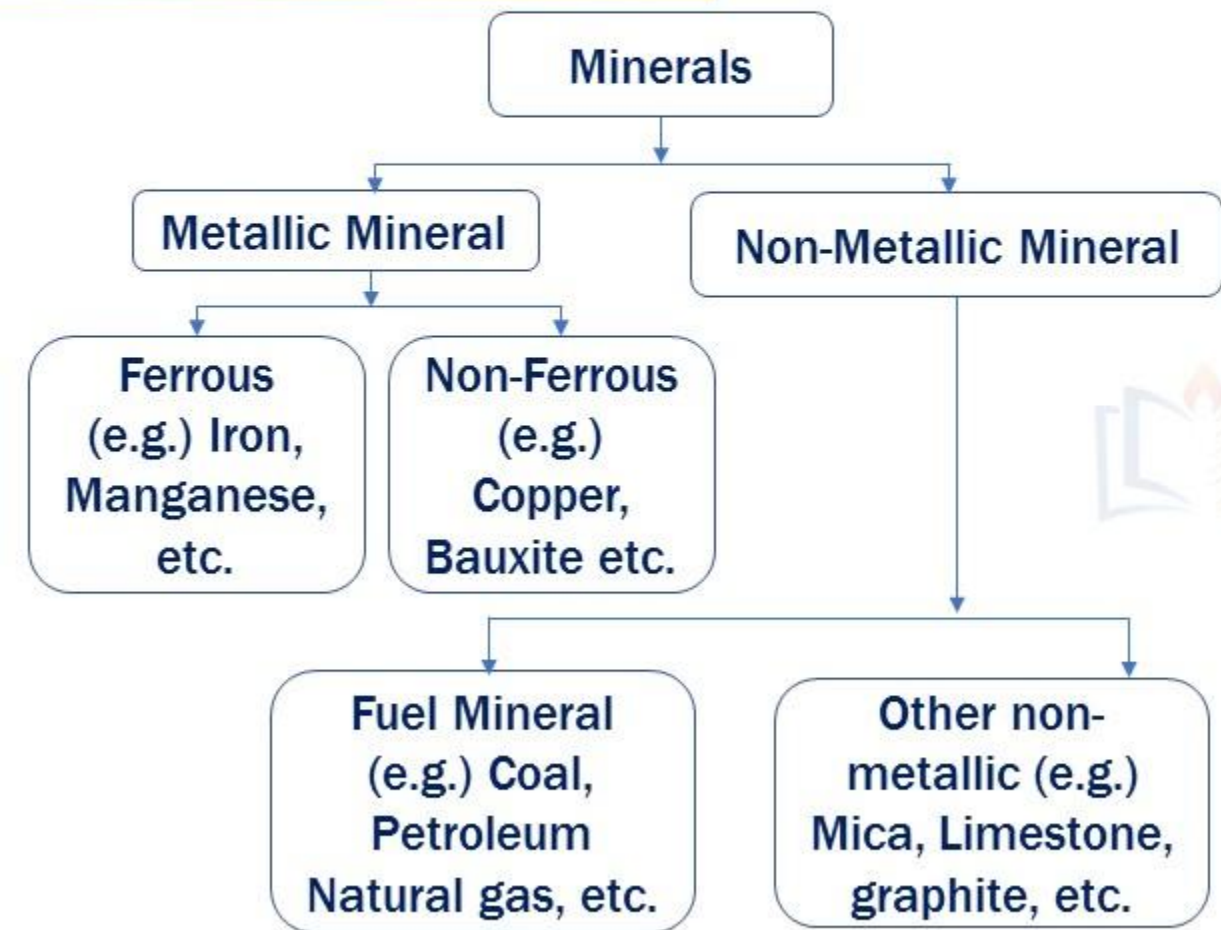
MODULE - 1

DISTRIBUTION AND TYPES OF MINERAL RESOURCES

MINERAL AND ENERGY RESOURCES

A mineral is a natural substance of organic or inorganic origin with definite chemical and physical properties.

Classification of Minerals



TYPES OF MINERAL RESOURCES

- 1 On the basis of chemical and physical properties, minerals may be grouped under two main categories of **metallic** and **non-metallics** which may further be classified as given here.
- 2 As it is clear from the chart, metallic minerals are the sources of metals.
- 3 Iron ore, copper, gold produce metal and are included in this category.

MINERAL AND ENERGY RESOURCES

TYPES OF MINERAL RESOURCES

METALLIC MINERALS

- 4 Metallic minerals are further divided into **ferrous and non-ferrous** metallic minerals. Ferrous, as you know, refers to iron.
- 5 All those minerals which have iron content are ferrous such as iron ore itself and those which do not have iron content are non-ferrous such as copper, bauxite, etc.

NON-METALLIC MINERALS

- 6 Non-metallic minerals are either **organic** in origin such as fossil fuels also known as mineral fuels which are derived from the buried animal and plant life such as coal and petroleum.
- 7 Other type of non-metallic minerals are **inorganic** in origin such as mica, limestone and graphite, etc.

MINERAL AND ENERGY RESOURCES

TYPES OF MINERAL RESOURCES

CERTAIN CHARACTERISTICS OF MINERALS

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Minerals have certain characteristics.

- These are **unevenly distributed** over space.
- There is **inverse relationship in quality and quantity** of minerals i.e. good quality minerals are less in quantity as compared to low quality minerals.
- The third main characteristic is that all minerals are **exhaustible** over time.
- These take long time to develop geologically and they cannot be replenished immediately at the time of need.
- Thus, they have to be conserved and not misused as they do not have the second crop.

AGENCIES INVOLVED IN THE EXPLORATION OF MINERALS



National Aluminium Company Limited

In India, systematic surveying, prospecting and exploration for minerals is undertaken by the :

- Geological Survey of India (GSI)
- Oil and Natural Gas Commission (ONGC)
- Mineral Exploration Corporation Ltd.(MECL)
- National Mineral Development Corporation (NMDC)
- Indian Bureau of Mines(IBM)
- Bharat Gold Mines Ltd. (BGML)
- Hindustan Copper Ltd. (HCL)
- National Aluminium Company Ltd. (NALCO) and;
- The Departments of Mining and Geology in various states.

MINERAL AND ENERGY RESOURCES

DISTRIBUTION OF MINERALS IN INDIA

- 1 Most of the metallic minerals in India occur in the Peninsular Plateau region in the old crystalline rocks.
- 2 Over 97 per cent of coal reserves occur in the valleys of Damodar, Son, Mahanadi and Godavari.
- 3 Petroleum reserves are located in the sedimentary basins of Assam, Gujarat and Mumbai High i.e. off-shore region in the Arabian Sea.
- 4 New reserves have been located in the Krishna-Godavari and Kaveri basins.
- 5 Minerals are generally concentrated in the following broad belts in India.
- 6 There may be some sporadic occurrences here and there in isolated pockets.



- A. North-Eastern Peninsular Belt**
- B. Central Belt**
- C. Southern Belt**
- D. South-western Belt**
- E. North-western Belt**

	MANGANESE
	BAUXITE
	OIL FIELDS
	LEAD
	COAL
	IRON ORE
	COPPER ORE
	SULPHUR
	MICA
	GOLD

MINERAL AND ENERGY RESOURCES

DISTRIBUTION OF MINERALS IN INDIA

1 THE NORTH-EASTERN PLATEAU REGION

1 This belt comprising of **Chota Nagpur plateau and Orissa Plateau** in the States of Jharkhand, West Bengal and Orissa is the **richest mineral belt of India**

2 It contains large quantities of coal, iron ore, manganese, mica, bauxite, copper, kyanite, chromite, beryl, apatite and many more minerals.

3 In fact you ask for any major mineral of India and you will find it in this belt.

4 Thus, it is a **mineral region par excellence**. The Chota Nagpur Plateau is known as the **mineral heart land of India**.

5 According to Wadia, this region possesses India's

- 100% Kyantie,
- 93% iron ore,
- 84% coal,
- 70% chromite,
- 70% mica,
- 45% Asbestos,
- 20% limestone and
- 10% manganese.

However, many changes have taken place in the recent years.

MINERAL AND ENERGY RESOURCES

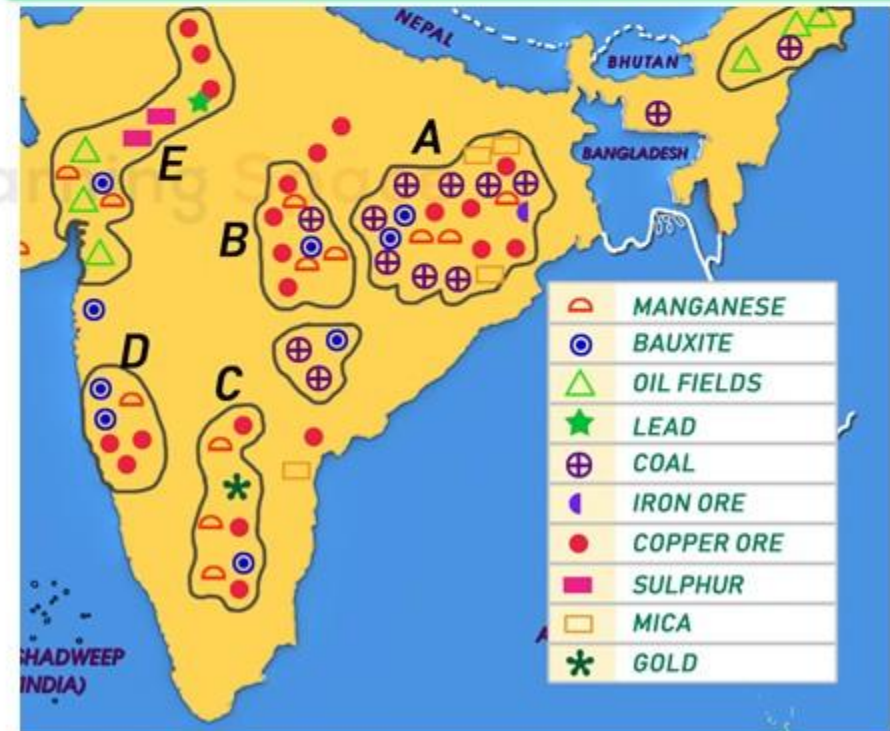
DISTRIBUTION OF MINERALS IN INDIA

2 CENTRAL BELT

- 1 This belt encompassing parts of Chhattisgarh, Madhya Pradesh, some parts of Telangana, Andhra Pradesh and Maharashtra is the second largest mineral belt of India.
- 2 Large deposits of manganese, bauxite, limestone, marble, coal, gems (Panna), mica, iron ore, graphite etc. are available here.
- 3 A comprehensive geological survey is still needed to know about the mineral wealth of this belt.

3 THE SOUTH-WESTERN BELT

Southern Karnataka and Goa are included in this belt. This belt is rich in manganese and bauxite.



MINERAL AND ENERGY RESOURCES

DISTRIBUTION OF MINERALS IN INDIA

4 THE SOUTHERN BELT



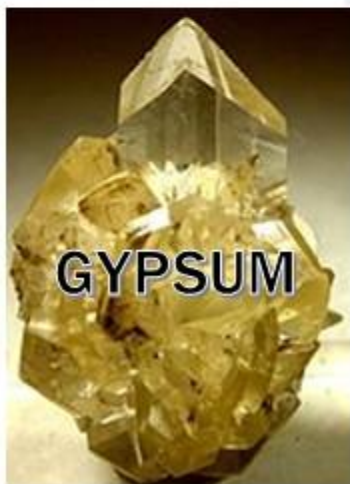
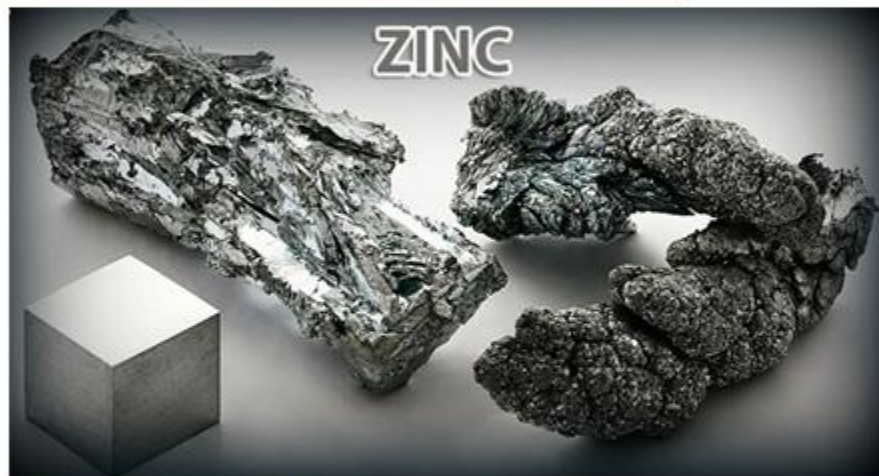
Neyveli Lignite Corporation

- 1 It covers mostly the Karnataka plateau but extends over the contiguous Tamil Nadu upland.
- 2 It is more or less similar to the north-eastern peninsular belt, as far as deposits of ferrous minerals and bauxite are concerned but it lacks coal deposits excepting lignite at Neyveli.
- 3 However, its mineral diversity is not as pronounced as that of the north-eastern peninsular belt.

MINERAL AND ENERGY RESOURCES

DISTRIBUTION OF MINERALS IN INDIA

5 THE NORTH-WESTERN BELT



- 1 This belt extends along the Aravalis in Rajasthan and in adjoining parts of Gujarat.
- 2 This belt has developed recently and is gradually becoming a productive region, holding great promise for the mining of metals like copper, lead, zinc, uranium, mica, steatite, beryllium and precious stones (aquamarine and emerald).
- 3 Gujarat is fast becoming an important producer, besides producing gypsum, manganese, salt, bauxite etc.

MINERAL AND ENERGY RESOURCES

DISTRIBUTION OF MINERALS IN INDIA

Outside the main belts described above, minerals in some other parts of the country are scattered here and there.

6 OTHER SPORADIC DISTRIBUTION SITES

- 1 **Assam** has reserves of petroleum and lignite.
- 2 The Himalayan region has some deposits of coal, bauxite, copper, slate etc.
- 3 **Mumbai High and Godavari basin** have reserves of oil and natural gas.

- 4 With the advancement of technology, even the **sea bed is being exploited for minerals**.
- 5 There are great possibilities of obtaining large quantities of oil and natural gas, uranium, manganese, copper, zinc, lead and so many other minerals from the sea bed.

1

IRON ORE

- 1 India is endowed with fairly abundant resources of iron ore.
- 2 It has the largest reserve of iron ore in Asia.



MAGNETITE



HAEMATITE

- 3 The two main types of ore found in our country are **haematite and magnetite**.
- 4 It has great demand in international market due to its superior quality.
- 5 The iron ore mines occur in **close proximity to the coal fields** in the north-eastern plateau region of the country which adds to their advantage.
- 6 About 95 per cent of total reserves of iron ore is located in the States of **Odisha, Jharkhand, Chhattisgarh, Karnataka, Goa, Telangana, Andhra Pradesh and Tamil Nadu**.

METALLIC MINERALS

IRON - ORE

1. HAEMATITE MINES



CHINA

PAKISTAN

ARABIA

OMAN

INDIA

NOAMUNDI MINES - Famous
(IRON ORE IS OF HIGHEST QUALITY)

GURUMAHISANI MINES

BADAMPAHAR MINES

BAILADILLA MINES

DALLI-RAJHARA MINES

KEMMANNUGUNDI MINES OF
BABABUDAN HILLS

KUDUREMUKHA MINES

KARNATAKA

CHHATTISGARH

JHARKHAND

ODISHA

MYANMAR
(BURMA)

Bay of Bengal

Arabian Sea

METALLIC MINERALS

IRON - ORE

1. HAEMATITE ORE



METALLIC MINERALS

IRON - ORE

2. MAGNETITE ORE



IRON ORE



7 In Odisha, iron ore occurs in a series of hill ranges in Sundergarh, Mayurbhanj and Keonjhar.

8 The important mines are Gurumahisani, Sulaipet, Badampahar (Mayurbhanj), Kiruburu(Kendujhar) and Bonai (Sundergarh).

9 Jharkhand has some of the oldest iron ore mines and most of the iron and steel plants are located around them.

10 Most of the important mines such as Noamundi and Gua are located in Poorbi and Pashchimi Singhbhum districts.

11 This belt further extends to Durg, Dantewara and Bailadila.

12 Dalli and Rajhara in Durg are the important mines of iron ore in the country.

METALLIC MINERALS

IRON - ORE

1. HAEMATITE ORE

13

In Karnataka, iron ore deposits occur in Sandur-Hospet area of Ballari district, Baba Budan hills and Kudremukh in Chikkamagaluru district and parts of Shivamogga, Chitradurg and Tumakuru districts.

Arabian Sea

GOA

KARNATAKA

TELANGANA

ANDHRA PRADESH

Bengal

KEMMANNUGUNDI MINES OF
BABABUDAN HILLS

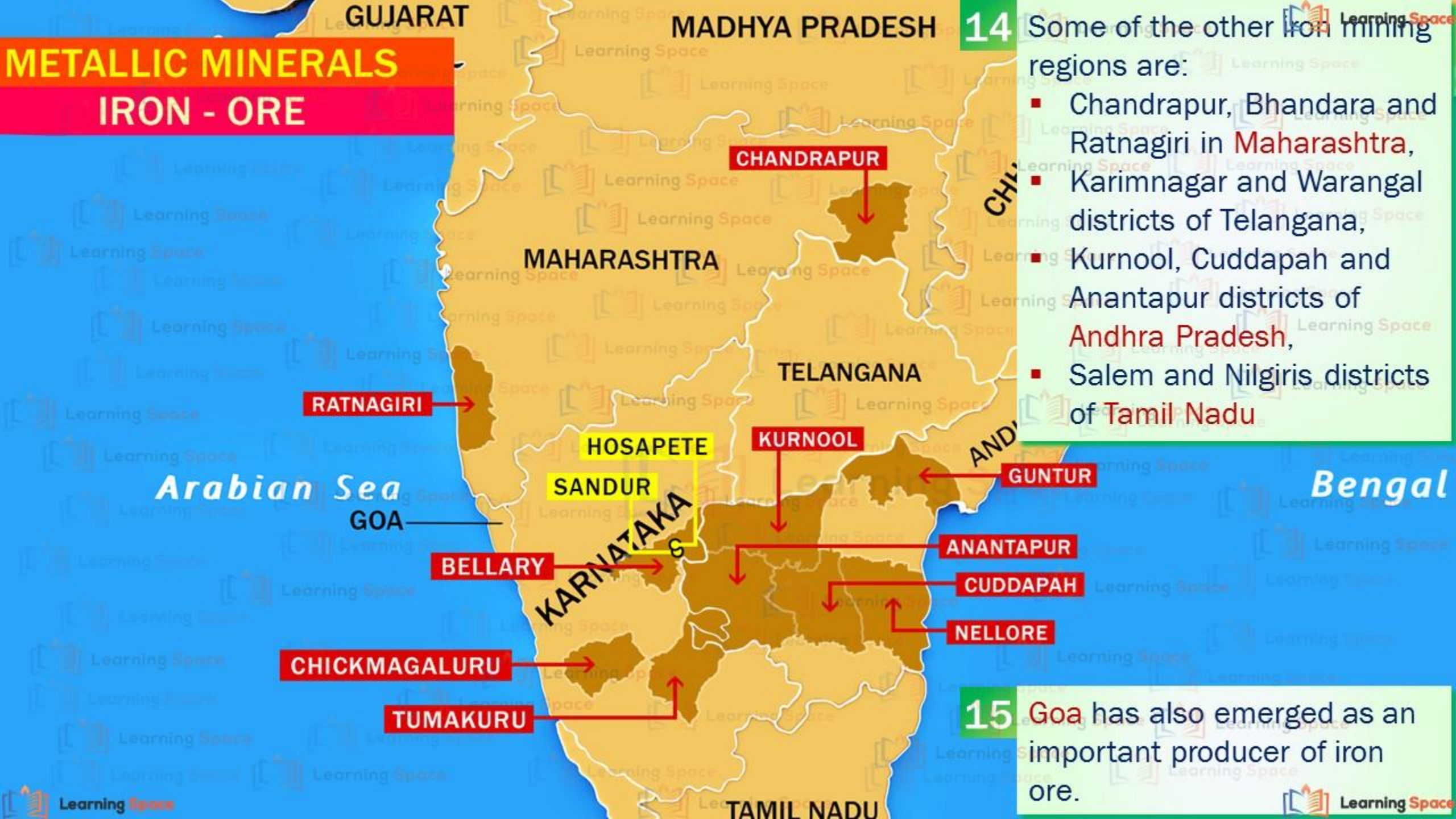
KUDUREMUKHA MINES

TAMIL NADU

KERALA

METALLIC MINERALS

IRON - ORE



MADHYA PRADESH

14

Some of the other iron mining regions are:

- Chandrapur, Bhandara and Ratnagiri in Maharashtra,
- Karimnagar and Warangal districts of Telangana,
- Kurnool, Cuddapah and Anantapur districts of Andhra Pradesh,
- Salem and Nilgiris districts of Tamil Nadu

Bengal

15

Goa has also emerged as an important producer of iron ore.

METALLIC MINERALS

IRON - ORE

3. LIMONITE ORE



2 MANGANESE

1 Manganese is an important raw material for **smelting of iron** and also used for **manufacturing ferro alloys**.

2 Manganese deposits are found in almost all geological formations, however, it is mainly associated with **Dharwar system**.

3 ODISHA

- It is the leading producer of Manganese.
- Major mines in Odisha are located in the central part of the iron ore belt of India, particularly in Bonai, Kendujhar, Sundergarh, Gangpur, Koraput, Kalahandi and Bolangir.

4 KARNATAKA

- Another major producer and here the mines are located in Dharwar, Ballari, Belagavi, North Canara, Chikkmagaluru, Shivamogga, Chitradurg and Tumkur.

5 MAHARASHTRA

- Mines in Nagpur, Bhandara and Ratnagiri districts.

6 MADHYA PRADESH

- Belt of mines in Balaghat-Chhindwara-Mandla and Jhabua districts.

7 Telangana, Goa, and Jharkhand are other minor producers of manganese.

METALLIC MINERALS

MANGANESE



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